

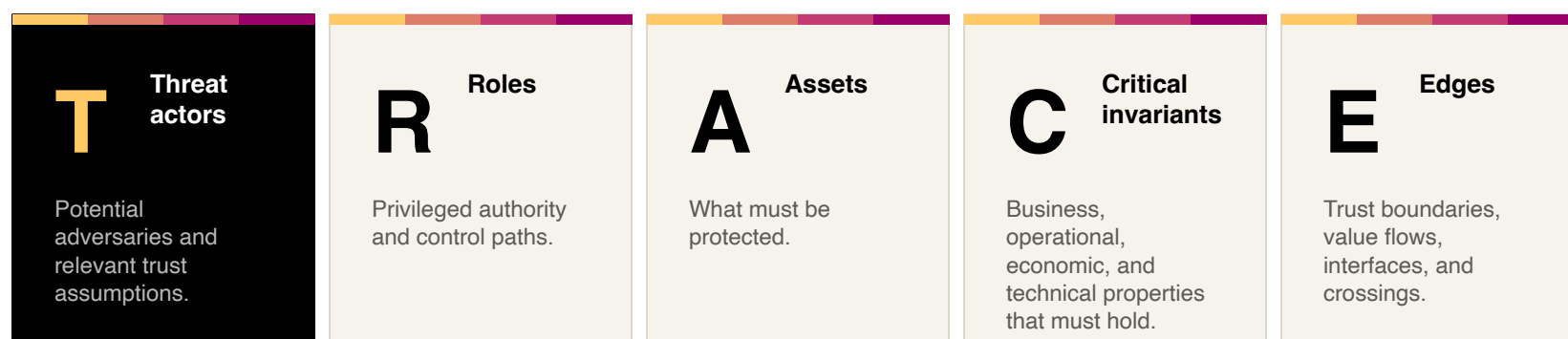
TRACE Threat Modeling Framework

A method for modern decentralized organisations that turns heterogeneous source material into a structured model, then uses STRIDE, attack trees, and optional coordination analysis to produce decision-ready security work.

Why modern organisations need TRACE

Modern cloud-first organisations combine high-value assets, distributed authority, SaaS and cloud systems, remote operations, and human control paths. TRACE keeps these in one model: STRIDE for conventional computing risk, plus attack trees and coordination analysis for protocol, system, and organisation risk.

TRACE model primitives



Inputs



Output

A traceable model that connects each material threat to an asset, role, invariant, trust boundary, flow, attack tree, and mitigation decision.

From source material to attack paths.

AI accelerates decomposition and coverage. Senior security experts own the assumptions, severity, plausibility, and recommendations.

01 Ingest sources

Docs, reviews, architecture, policies, code, configs, and organisational context.

02 Construct model

Assets, actors, invariants, boundaries, value flows, authority paths, roles, components.

03 STRIDE analysis

Structured threat pass across components, flows, and privilege boundaries.

04 Attack trees

Top threats decomposed into plausible exploit paths and enabling conditions.

05 Coordination inspection

Optional pass for governance bodies, committees, vendors, administrators, operators, and other actor groups.

06 Roadmap outputs

Risk ranking, mitigations, open questions, and assessment, audit, or launch guidance.

Designed with human-AI co-working in mind

A deliberate sequential workflow uses AI for decomposition, coverage, and draft structure while senior security experts own the risk decisions.

- AI proposes source inventories, model candidates, and coverage gaps.
- Experts approve and complement assumptions, scope, and relevance.
- Experts set severity, plausibility, and remediation guidance.

Human Decision Points

Source inventory, TRACE model, STRIDE Threat identification and ranking, attack trees, roadmap.